

Solutions to TECH3 Exam 2026

Problem 1

- a) Never use a pie chart. Ever. There are limitation to how we humans perceive and are able to compare different angles and areas/volumes. The figure shows that the female board member proportion is rather stable (around one fifth to a quarter) for the period studied, but its almost impossible to detect any particular trend over time by looking at the pie charts. The data is actually a proportion, but it would be better to simply plot the percentage of female board members on the y-axis and the years on the x-axis, either as a bar plot or a line plot (as the figure in b).

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6 points for discussion of graphic and 4 points for suggesting a better way of showing the same data. A good answer should include something along the lines of “Never use a pie chart. Ever.”. Note that students may detect other issues with the graphs and suggest other improvements, and should be awarded for that.

- b) There are two main problems with the graphic:

- Data/ink-ratio is low: The grid lines take too much attention from the data, i.e. the data/ink-ratio is very low. If you want grid lines, the color should be less prominent.
- Dual y-axis: One should also never use dual y-axis. It easily creates misconceptions and makes the graphic hard to read. Which line should I interpret on which axis. Also, the grid lines are not aligned with the values on the righthand axis.

It would be better to use two panels - one for each time series and remove the grid lines completely or make them less prominent.

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3 points for pointing out each of the bullet points, and 4 points for suggesting a better way of showing the same data. Note that students may detect other issues with the graphs and suggest other improvements, and should be awarded for that.

- c) The claim: According to the model, an increase of one percentage point of female board members is **associated with** an increase in global mean temperature of 0.12°C , but there is no causal relation.

The plot indicates that global mean temperature anomalies has been increasing over the period 2007-2023. The same is true for the proportion of female board members. The graphic is made to tell the message that the increased proportion of female board members is contributing to climate change. What the graphic is actually showing is that both have had a increasing trend over this time period. We could consider *time* as a confounding variable in the sense that it affects both proportion of female board members and global temperature anomalies, even though the two variables are unrelated.

The regression is an example of a spurious regression. The regression coefficient is significantly different from zero at a 5% significance level, but there is no real association between the y- and x-variable - both a just increasing with time.

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3 points for commenting on the claim, 3 points for pointing out that the regression is spurious and 4 points for a discussion of common trend and year as a **confounding variable**(or confounder) between share of female board members and global temperature anomalies.

- d) The model states that $\widehat{\text{Temp}} = -2.0286 + 0.1156 \cdot \text{propFemale}$. The predicted temperature anomaly with zero female board members is therefore -2.0286°C . With 100% female board members (zero male), the expected anomaly is

$$-2.0286 + 0.1156 \cdot 100 = 9.5314,$$

i.e. 9.5°C . Note that both these values are extrapolations - far outside the range of female board member proportions the model was trained on.

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4 points for each prediction. 2 points for mentioning that these are extrapolations.

Problem 2

- a) Setting up the maximum likelihood estimator:

$$L(\lambda) = \prod_{i=1}^n f(x_i) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n e^{-\lambda \sum_{i=1}^n x_i}$$

Log-likelihood function:

$$\ell(\lambda) = \log L(\lambda) = n \log \lambda - \lambda \sum_{i=1}^n x_i$$

Setting the derivative to zero and solving for λ :

$$\frac{\partial \ell(\lambda)}{\partial \lambda} = \frac{n}{\lambda} - \sum_{i=1}^n x_i = 0$$

$$\frac{n}{\lambda} = \sum_{i=1}^n x_i \iff \hat{\lambda} = \frac{n}{\sum_{i=1}^n x_i}.$$

The second derivative with respect to λ is

$$\frac{\partial^2 \ell}{\partial \lambda^2} = \frac{-n}{\lambda^2} < 0,$$

indicating that the log-likelihood is a concave function of λ and that $\hat{\lambda}$ is a maximum, i.e. the maximum likelihood estimator.

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10 points for deriving the expression by going through the steps. Note that students do not need to use math-symbols, so sensors should be tolerant with the math syntax being used.

- b) Since $E(\hat{\lambda}) \neq \lambda$ the estimator is biased (not unbiased). However, when $n \rightarrow \infty$, the bias goes to zero, i.e.

$$\lim_{n \rightarrow \infty} E(\hat{\lambda}) = \lim_{n \rightarrow \infty} \frac{n}{n-1} \lambda = \lambda,$$

hence the MLE is asymptotically unbiased.

When $n \rightarrow \infty$ the variance will go to zero. Since, the bias also goes to zero when n becomes large, the estimator is consistent.

$$\lim_{n \rightarrow \infty} \text{Var}(\hat{\lambda}) = \lim_{n \rightarrow \infty} \frac{n^2 \lambda^2}{(n-1)^2 (n-2)} \left[\frac{O(n^2)}{O(n^3)} \right] = 0.$$

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4 points for biased and 6 points for consistent due to asymptotically unbiased.

c)

```
import numpy as np # Code A
time_between_purchases = np.array(
    [0.5, 0.1, 2.7, 0.4, 8.9,
     1.2, 0.6, 1.8, 7.4, 6.2,
     0.9, 1.3, 0.6, 6.8, 3.6])
lambdahat = 1/time_between_purchases.mean()
print(f"MLE: {lambdahat:.3f}")
```

MLE: 0.349

$\hat{\lambda} = 0.349$. This means that the average customer makes 0.349 purchases per day, or equivalently the average time between purchases is approximately $1/0.349 = 2.87$ days.

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5 points for calculating the estimate and 5 points for correct interpretation.

- d) Using the formula, and inserting the values provided, we get the formula asked for:

$$\mu(p) = p E(U) + (1 - p)E(X) = p \cdot 0.5 + (1 - p) \cdot \frac{1}{\lambda} = \frac{1}{\lambda} + \left(\frac{1}{2} - \frac{1}{\lambda}\right)p.$$

The derivative of μ is

$$\mu'(p) = \frac{1}{2} - \frac{1}{\lambda} = \frac{\lambda - 2}{2\lambda},$$

which is positive if $\lambda > 2$ (increasing function of p), and decreasing if $\lambda < 2$. If $\lambda = 2$, the function is a constant in terms of p . This is because when $\lambda = 2$, $E(X) = E(U) = E(Y)$. With $\lambda = 0.35$, we get

$$\mu(0.3) = \frac{1}{0.35} + \left(\frac{1}{2} - \frac{1}{0.35}\right)0.3 = 2.15.$$

We need to compare this number to $E(X) = 1/\lambda = 1/0.35 = 2.86$. Since $E(Y) < E(X)$, the expected time between purchases is lower with the campaign.

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3 points for finding the expression, 4 points for deriving the intervals and 3 points for comparison with and without the campaign.

e)

```
from scipy import stats # Code B
import numpy as np
def simulate_y(p=0.1, lam = 0.5, n=1) :
    u = stats.uniform.rvs(size = n)
    x = stats.expon.rvs(size = n, loc= 0, scale = 1/lam)
    react = stats.binom.rvs(size = n, n = 1, p=p)
    y = react*u+(1-react)*x
    return y
# Run one simulation:
np.random.seed(1234)
y = simulate_y(p = 0.3, lam = 0.35, n = 10000)
print(f"Mean: {y.mean():.2f}\nVariance: {y.var():.2f}")
```

Mean: 2.17

Variance: 7.29

The mean is estimated to 2.17 and variance 7.34.

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This exercise must be done using Python. Note that they just need to use the function. If they have made a for-loop, or used any other way of repeating the function, we do not have any objections to that. 4 points for being able to set up the Monte Carlo simulation and 3 points for each of the numbers.

f) Since we are provided with the mean (2.2) and variance of Y (7.3), we use that

$$\bar{Y}_{25} \sim N(E(Y), \frac{\text{Var}(Y)}{25}),$$

such that

```
from scipy.stats import norm
#P(meanY<1):
prob = norm.cdf(1, loc = 2.2, scale=np.sqrt(7.3/25))
print(f"Probability of mean of Y < 1: {prob:.4f}")
```

Probability of mean of Y < 1: 0.0132

A potential issue with this approximation is that the time between purchases has to be positive, so the time between purchases is truncated at zero, while the normal distribution covers $(-\infty, \infty)$. If the mean is high and variance is low, it may however be a good approximation, since then negative values in the approximation will be extremely unlikely.

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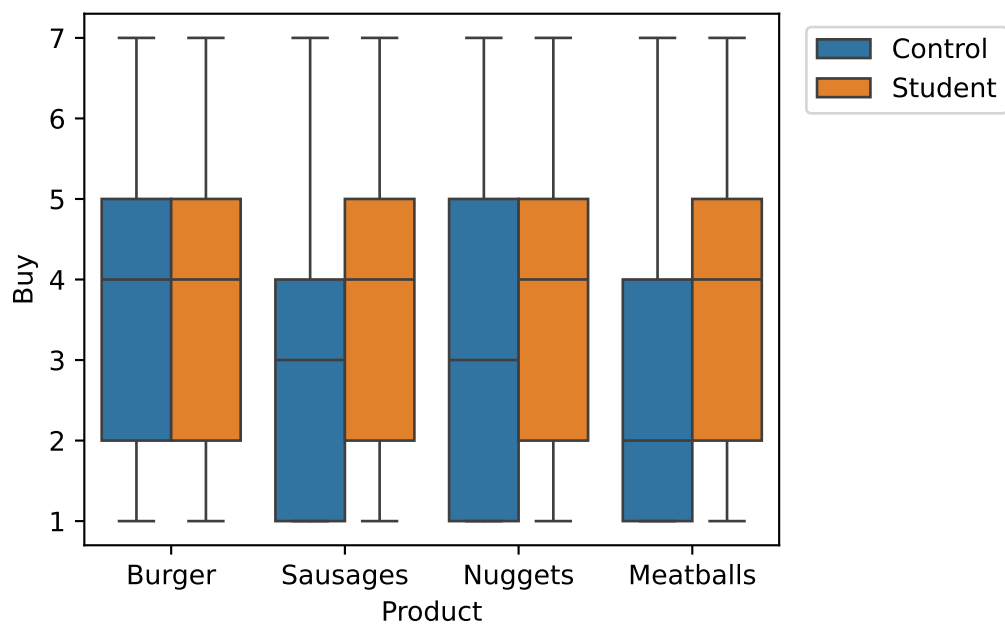
6 points for calculation and 4 points for discussing problems with it.

Problem 3

```
import pandas as pd # Code C
data = pd.read_csv("mab2experiment.csv", sep = ";")
data.head(5)
```

a)

```
import seaborn as sns
import matplotlib.pyplot as plt
sns.boxplot(x = "Product", y = "Buy", hue = "Group", data = data)
plt.legend(bbox_to_anchor=(1.02, 1), loc="upper left")
plt.tight_layout()
plt.show()
```



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10 points for making nice graphic(s). Some student may make it using `hue = "Group"` argument, but if students choose to use four different plots, we do not subtract points for that.

b) Hypotheses:

- H_0 : Likelihood to buy and Group (student vs control) are independent variables.
- H_1 : Likelihood to buy and Group (student vs control) are dependent variables.

```
product_df = data.loc[data["Product"]=="Meatballs"]
contingency_table = pd.crosstab(                                # Code E
    index=product_df["Buy"],
    columns=product_df["Group"]
)
print(contingency_table)
```

Group	Control	Student
Buy		
1	74	33
2	42	22
3	36	34
4	36	28
5	12	48
6	5	34
7	4	9

```
from scipy.stats import chi2_contingency
stat, p, dof, expected = chi2_contingency(contingency_table)
print(f"Stat: {stat:.2f}\n"
      f"p-value: {p:.5f}")
```

Stat: 68.10
p-value: 0.00000

The test statistic is high and the p-value is close to zero. We clearly reject the null hypothesis of independence. The person's likelihood to buy the meatballs is not independent of which ads the person is exposed to.

If a student does the exercise by hand, they get full score. The expected frequencies are then:

```
expected_df = pd.DataFrame(
    np.round(expected, 2),
    index=contingency_table.index,
    columns=contingency_table.columns
)
print(expected_df)
```

Group	Control	Student
Buy		
1	53.63	53.37
2	32.08	31.92
3	35.08	34.92
4	32.08	31.92
5	30.07	29.93
6	19.55	19.45
7	6.52	6.48

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3 points for setting up the hypothesis, 4 points for calculating test statistic and p-value (either “by hand” or using python) and 3 points for conclusion based on this. Note that students may also just use the p-values as a measure of how much we should believe in the null hypothesis and not make a firm conclusion. In this particular exercise, the p-value is zero, so the evidence is strong against the null.

- c) Let μ_c and μ_s denote the mean purchase likelihood scores for the control and student groups, respectively. We then want to test if

$$H_0 : \mu_s = \mu_c \quad H_A : \mu_s > \mu_c.$$

Here we may use $\alpha = 5\%$, as there is not really any large concerns with making type I or II errors.

```
student = product_df.loc[product_df.Group=="Student"]["Buy"]
control = product_df.loc[product_df.Group=="Control"]["Buy"]
stat, p = stats.ttest_ind(student, control, alternative='greater')
print(f"stat: {stat :.2f}")
print(f"p-value: {p :.7f}")
```

```
stat: 7.99
p-value: 0.0000000
```

The p-value is zero also for this test. The evidence suggests that the people exposed to the student’s ad has a higher expected likelihood to buy the meatballs, compared to the group exposed to the control ads. This test is directional, so we can conclude that the effect goes in the direction of student’s ad having a higher effect than the control ad, while the chi-square test in (b) only tests if there is a relation or not.

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3 points for setting up the hypothesis, 3 points for calculating test statistic and p-value (either “by hand” or using python) and 2 points for conclusion based on this. Note that students may also just use the p-values as a measure of how much we should believe in the null hypothesis and not make a firm conclusion. In this particular exercise, the p-value is zero, so the evidence is strong against the null. 2 points for discussion of directional test vs non-directional.

d)

```
from statsmodels.formula.api import ols
mod = ols(formula = "Buy ~ Group", data = product_df)
fit = mod.fit()
print(fit.summary())
```

```

                                OLS Regression Results
=====
Dep. Variable:                  Buy    R-squared:                0.133
Model:                            OLS    Adj. R-squared:           0.131
Method:                 Least Squares    F-statistic:                63.89
Date:                Wed, 03 Jun 2026    Prob (F-statistic):          1.32e-14
Time:                  13:00:45    Log-Likelihood:             -805.47
No. Observations:                417    AIC:                       1615.
Df Residuals:                    415    BIC:                       1623.
Df Model:                            1
Covariance Type:                nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	2.5263	0.116	21.821	0.000	2.299	2.754
Group[T.Student]	1.3102	0.164	7.993	0.000	0.988	1.632

```

=====
Omnibus:                21.727    Durbin-Watson:           2.114
Prob(Omnibus):           0.000    Jarque-Bera (JB):         11.098
Skew:                    0.203    Prob(JB):                 0.00389
Kurtosis:                2.312    Cond. No.:                2.62
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

From the printed summary, we see that the group effect (i.e. the effect of being exposed to the student's ad) is 1.3102. The test statistic is 7.993 and the p-value is 0. This test is almost the same as the test we did above, except that here the alternative hypothesis is two-sided, i.e. we test $H_0 : \beta_{\text{group}} = 0$ against $H_A : \beta_{\text{group}} \neq 0$. The conclusion here is that the coefficient associated with group is significantly different from zero at any reasonable significance level, while above we concluded that the effect was positive. Note that the test-statistic for the group effect here is exactly the same value as the test statistic in (c), it is only the alternative hypotheses that are different.

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5 points for producing the model summary and 3 points for conducting the test for the coefficient. 2 points for the link to the previous test.

e)

```

from statsmodels.formula.api import ols                                # Code F
fit1 = ols("Buy ~ Group + Age + Gender + Education + Income + Politics",
           data = product_df).fit()
fit2 = ols("Buy ~ Group + Gender + Education + Income + Politics",
           data = product_df).fit()
fit3 = ols("Buy ~ Group + Education + Politics",
           data=product_df).fit()
fit4 = ols("Buy ~ Group + Politics",
           data=product_df).fit()
print(f"Model 1: AIC {fit1.aic:.1f}\n"
      f"Model 2: AIC {fit2.aic:.1f}\n"
      f"Model 3: AIC {fit3.aic:.1f}\n"
      f"Model 4: AIC {fit4.aic:.1f}\n")

```

```

Model 1: AIC 1588.3
Model 2: AIC 1586.6
Model 3: AIC 1584.0
Model 4: AIC 1587.3

```

We select the model that minimizes AIC, in this case, this is the third model (*fit3*).

```
print(fit3.summary())
```

```

                                OLS Regression Results
=====
Dep. Variable:                  Buy    R-squared:                  0.203
Model:                          OLS    Adj. R-squared:             0.197
Method:                        Least Squares    F-statistic:                35.06
Date:                          Wed, 03 Jun 2026    Prob (F-statistic):        3.34e-20
Time:                          13:00:45    Log-Likelihood:            -788.02
No. Observations:                417    AIC:                        1584.
Df Residuals:                    413    BIC:                        1600.
Df Model:                        3
Covariance Type:                  nonrobust
=====

```

	coef	std err	t	P> t	[0.025	0.975]
Intercept	2.6800	0.478	5.609	0.000	1.741	3.619
Group[T.Student]	1.2994	0.158	8.206	0.000	0.988	1.611
Education	0.1348	0.059	2.285	0.023	0.019	0.251
Politics	-0.2776	0.056	-4.930	0.000	-0.388	-0.167

```

=====
Omnibus:                        5.561    Durbin-Watson:                2.137
Prob(Omnibus):                  0.062    Jarque-Bera (JB):              5.622
Skew:                           0.262    Prob(JB):                      0.0601
Kurtosis:                      2.781    Cond. No.                      45.4
=====

```

Notes:

[1] Standard Errors assume that the covariance matrix of the errors is correctly specified.

The model fitted here uses education and politics as continuous variables (even though they are strictly ordinal). We can write the model as

$$\text{Buy}_i = \beta_0 + \beta_1 \text{Student}_i + \beta_2 \text{Education}_i + \beta_3 \text{Politics}_i + \epsilon_i,$$

Inserting the estimated coefficients from the model summary:

$$\text{Buy}_i = 2.68 + 1.2994 \cdot \text{Student}_i + 0.1348 \cdot \text{Education}_i - 0.2776 \cdot \text{Politics}_i + \epsilon_i,$$

where Student_i is a dummy variable being one if person i is exposed the student's ad and zero otherwise. ϵ has expectation zero and constant variance $\hat{\sigma}^2 = 2.59$, which we can estimate using

```
print(f"Sigma2 = {fit3.mse_resid:.2f}")
```

Sigma2 = 2.59

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3 points for model selection by AIC and 7 points for setting up the equation with the 5 correct coefficients. Subtract points for minor mistakes. Reserve 2 points for commenting epsilon.

- f) The fitted versus residuals plot looks a bit strange - all the points lie on parallel lines. The two colored parallel lines indicate the fact that the variable $y = \text{Buy}$ is on a integer scale from 1-7, while $\hat{y} = \widehat{\text{Buy}}$ is continuous. The means that for all instances of $y = 1$, the residuals are $e = 1 - \hat{y}$, which, when plotted against $\hat{y}$, will be points on the line: $y = 1 - x$. This will be the same for $y = 2, 3, \dots, 7$ and is why we see points on parallel lines in the figure (orange/top line is $y = 7 - x$ and purple/lower line indicating $y = 1 - x$).

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6 points for the general explanation of the pattern and 4 points connecting the colored parallel lines to specific values of Y (1 and 7).